

Schema Guardrails in the Wild

Empirically Testing Whether Structural Constraints Tame LLM Output Volatility

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Same Input, Different Output

Run 1

```
{ "country":  
  "Mexico",  
  "risk_level": 2,  
  "event_types": [  
    "crime",  
    "kidnapping" ] }
```

Run 2

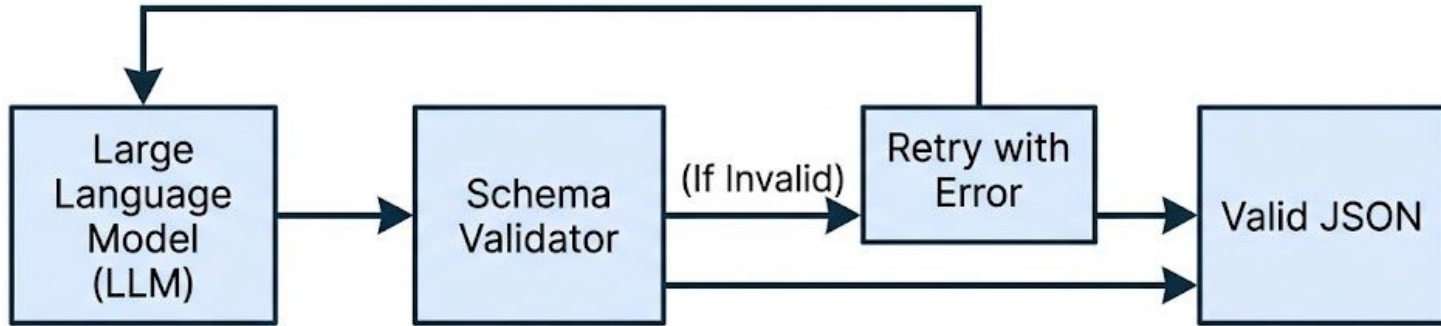
```
{ "country":  
  "Mexico",  
  "risk_level": 2,  
  "event_types": [  
    "crime",  
    "civil_unrest",  
    "other" ] }
```

Same model. Same prompt. Same input. Different output.

Downstream systems can't tell a model resampling artifact from a real change in the world.

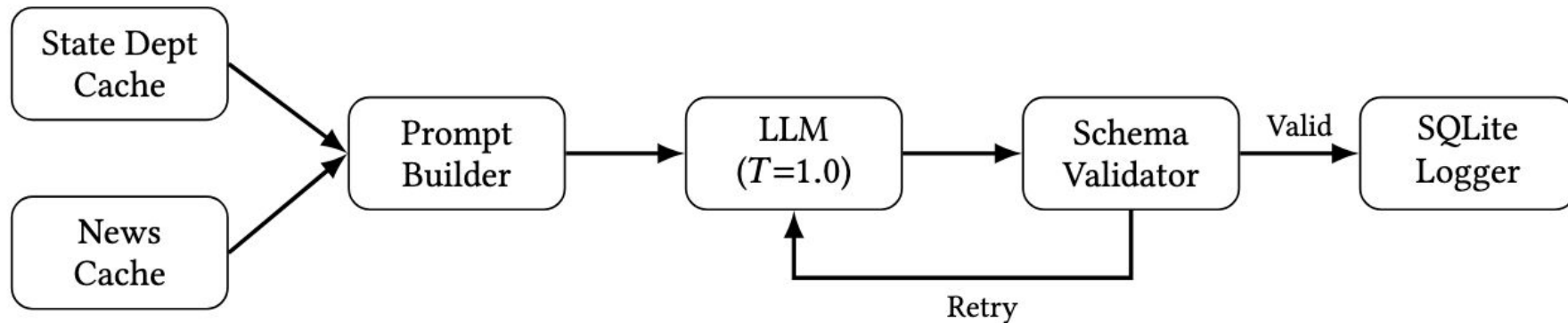
**Does applying schema constraints with
a validation-guided retry loop reduce
LLM output volatility by at least 50% in
a real-world structured IE task?**

The Standard Fix



- Catches malformed JSON
- Feeds validator's error back to model
- Marketed as a path to deterministic output

Travel Risk Assessment Pipeline



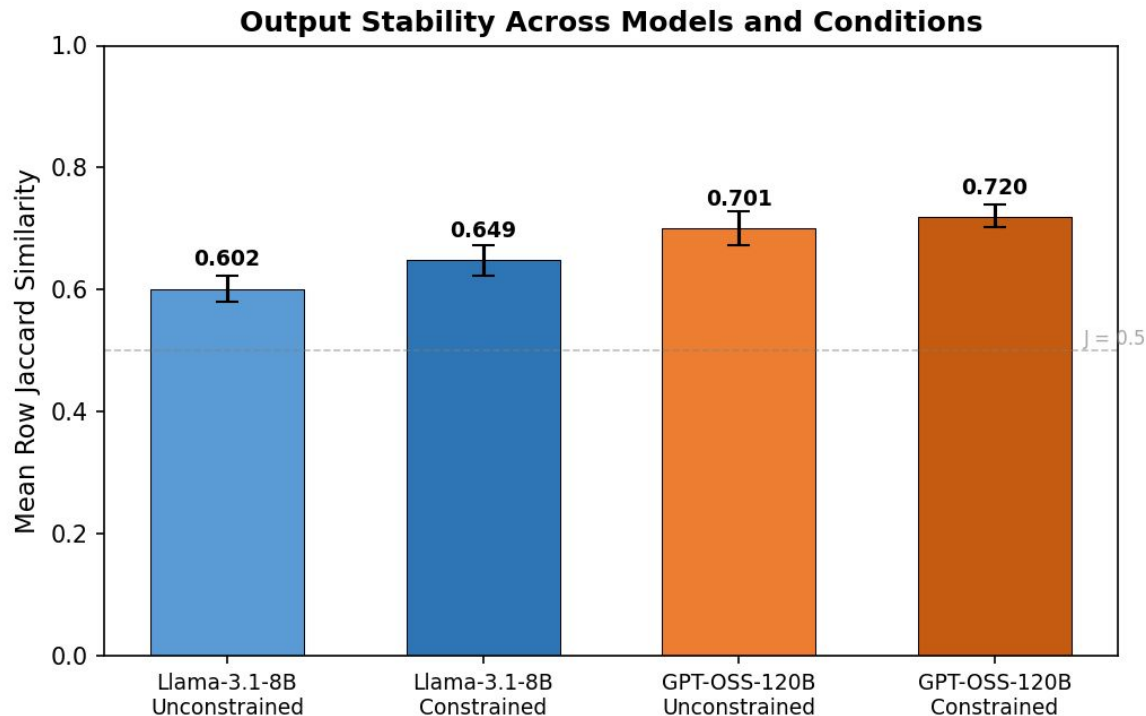
Experimental Setup:

- 30 countries, 5 runs each, 600 inference calls per condition
- Two models: gpt-oss-120b LLM vs. llama-3.1-8b-instruct SLM
- Two conditions: unconstrained vs. validate-and-retry
- All inputs cached: every bit of output variation is model behavior

Live Demo

<https://youtu.be/ue2nyMla8qQ>

Three Findings



1. Hypothesis rejected. 50% target. Actual: 11.9% (SLM), 4.8% (LLM, not significant).

2. Larger model wins. The 120B LLM is 26–29% more stable than the 8B SLM — in every condition.

3. Retries aren't free. +30–40% token overhead for the SLM's 11.9% gain.

Why Validators Can't Fix This

"Arauca" $\approx$ "Arauca department" $\approx$ "Arauca Department, Colombia"

"FATA region" $\approx$ "Federally Administered Tribal Areas" $\approx$ "tribal border areas"

"Northern Border" $\approx$ "northern border states" $\approx$ "Mexico-US border region"

Same place. Different string. Jaccard counts them as different

Takeaways

- **Bullet 1: Schema validation is a safety net, not a stability solution.** It catches malformed JSON, but structural rules can't enforce meaning.
- **Bullet 2: Smaller models can be more consistent than larger ones.** The 8B SLM was 26–29% more stable than the 120B LLM across every condition.
- **Bullet 3: The real fix is semantic, not structural.** Entity linking, canonicalization maps, and controlled vocabularies close the gap that validators can't.

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LEADING THE CHARGE, CHARGING AHEAD